Heat Pump Lab

ENGN 0720: Thermodynamics

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Abstract

The objective of the Air-Source Heat Pump laboratory was to investigate the operation of a vapor-compression heat pump cycle and to analyze its performance using the First Law of Thermodynamics. Heat pump technology in this experiment allows us to transport heat from a low-temperature evaporator to a high-temperature condenser, demonstrating how electrical input of work can provide more heat output than resistance heating can provide alone. With R134a as the working fluid, we applied the First Law of Thermodynamics as a closed system on the cycle. We measured significant performance parameters, including refrigerant pressures, temperatures, and coefficient of performance (COP). Measurements include a high-side pressure of about 129 psig, a low-side pressure of about 37 psig, and a compressor exit temperature of about 119 °F. The COP was calculated at about 3.0, meaning about three times as much heat energy was provided by the heat pump than what was used by the compressor.

Introduction

A heat pump refers to a space-saving and efficient thermodynamic heat-transfer device from a cold to a hot reservoir under very close to steady-flow conditions. In this laboratory, a North Park Innovations TU-807 table-top heat pump trainer charged with R134a refrigerant was used to measure and observe prominent thermodynamic characteristics of a vapor-compression cycle. Pressure and temperature probes were mounted on this loop to measure all sections through the compressor, condenser, evaporator, and expansion valve. Hence, we were able to compute energy changes in each of these stages accordingly.

With airflow and refrigerant pressure/temperature measurements, we could compute the heat added to and dumped by the working fluid in the evaporator and condenser. The measured data also allowed us to approximate the coefficient of performance (COP), an index of the heat pump's ability to deliver more heat output than electrical input. Because the selected system is a closed vapor compression with zero net refrigerant volume change on a cycle-average basis, we used The First Law of Thermodynamics in a form applicable to a device operating under a steady-flow process to define device performance:

$$\Delta H_{sys} = Q_{in} - Q_{out} + W_{in}$$

This laboratory demonstrated both ends of heat pump usage, from home air-conditioning and heating to commercial refrigeration. It also presented the heat pumps' energy-saving advantage over traditional electric resistance heating.

Experimental Procedure

At the start of each test, the tabletop heat pump trainer was brought to room temperature standard laboratory condition with its compressor de-activated and all its components having a few minutes of equilibrating time. With forced flow over the evaporator and condenser coils, all fans operated with their motor rotating at moderate speed. Thermocouple probes and pressure probes were tapped to convenient positions in the loop, i.e., high-pressure compressor discharge line and low-pressure suction line back to the compressor, as well as each coil's air inlet and outlet. We first gathered data in the system's "Heat" mode, and its reversing valve was energized to circulate refrigerant in heating mode. The compressor running speed was reduced to about 80% of its maximum to establish a clean pressure differential over its complete cycle without compromising excessive oscillations. A handheld anemometer measured airflow near each intake fan, and thereby, the volumetric flow rate was calculated, and clamp-on thermocouple probes measured line temperatures of refrigerant. Free-hanging probes monitored air temperature and relative humidity upstream and downstream of both coils.

Following leveling of readings—typically evident by stabilized pressures and negligible fluctuation in measured temperatures—readings were taken, e.g., high- and low-side pressures, compressor discharge line temperature and inlet line temperature, air temperature, and change in humidity across evaporator and condenser, and sight glass indications to see if anything of the two-phase mixture was visible or not visible. Note was taken if a change in temperature in refrigerant was noted between compression, condensing, expanding, and evaporating phases. After these measurements, the system stem was switched briefly into "Cool," so reversing the valve was reversing coil operations accordingly, so what was initially condenser became evaporator, and vice versa. It was briefly possible to see how this physical assembly created cooling compared to heating. After these measurements, the system was shut down. All data taken—pressure, temperatures, relative humidity, and air velocity—were employed to construct a T–S diagram of the R134a cycle, estimate refrigerant mass flow rate based on air-side enthalpy difference, and compute the coefficient of performance (COP) by applying First Law of Thermodynamics to measured energy fluxes.

Results

Table 1: Pressure Readings For Hot & Cold Cycle (PSIG)

	High Pressure in Cycle	Low Pressure in Cycle
Hot	131	42.2
Cold	128.6	36.9

Table 2: Temperatures at Cycle Stages (° F)

	Temp at Evaporator Outlet	Temp at Condenser Outlet	Temp at Compressor Outlet
Hot	51.5	100	120
Cold	97.2	45.9	119

Table 3: Air Properties of the Evaporator

	Supply Air		*Return Air*	
Evaporator	Temperature (° F)	Relative Humidity	Temperature (° F)	Relative Humidity
Hot	75.3	28.6	54.8	50.7
Cold	75.5	27.5	96.6	15.6

Table 4: Air Properties of the Condenser

	Supply Air		*Return Air*	
Condenser	Temperature (° F)	Relative Humidity	Temperature (° F)	Relative Humidity
Hot	72.1	31.1	100.8	13.8
Cold	72.6	30.5	54.1	52.1

Table 5: Air Speed Measurements (ft/min)

Evaporator	**In**	**Out**
Hot	669.3	492.1
Cold	669.3	433.1
Condenser		
Hot	708.7	393.7
Cold	689	393.7

Table 6: Calculated Thermodynamic Properties at each stage

Evaporator Outlet	Specific Entropy (Btu/lbm°R)	Specific Enthalpy (Btu/lbm)	State (% Quality if applicable)
Hot	0.22470	111.40	Superheated Vapor
Cold	0.09002	44.13	Subcooled Liquid
Compressor Outlet			
Hot	0.22650	121.00	Superheated Vapor
Cold	0.22700	121.10	Superheated Vapor
Condenser Outlet			
Hot	0.09178	45.12	Subcooled Liquid
Cold	0.22350	109.90	Superheated Vapor
Expansion Valve Outlet			
Hot	0.09394	45.12	Saturated Mixture (21.75%)
Cold	0.09237	44.13	Saturated Mixture (27.75%)

Discussion

The collected data reflects the performance of a vapor-compression heat pump cycle operating in heating mode. We established key thermodynamic states of the R134a working fluid by measuring its temperature and pressure at critical points in the system, then used an R134a property table to estimate its enthalpy and entropy at these points.

The low-side pressure in the cycle was found to be 42.2 psig and temperature at the evaporator outlet was 51.5°F, suggesting that the refrigerant exiting the evaporator was in a superheated vapor state, ready for compression. After compression, the high-side pressure in the cycle was found to be 131 psig, and the temperature at the compressor outlet reached 120°F, confirming significant superheating due to compression driven by electrical work input.

At the condenser outlet, the refrigerant temperature dropped to 100°F while remaining at a high pressure, indicating it transitioned to a subcooled liquid state. Continuing, the refrigerant entered

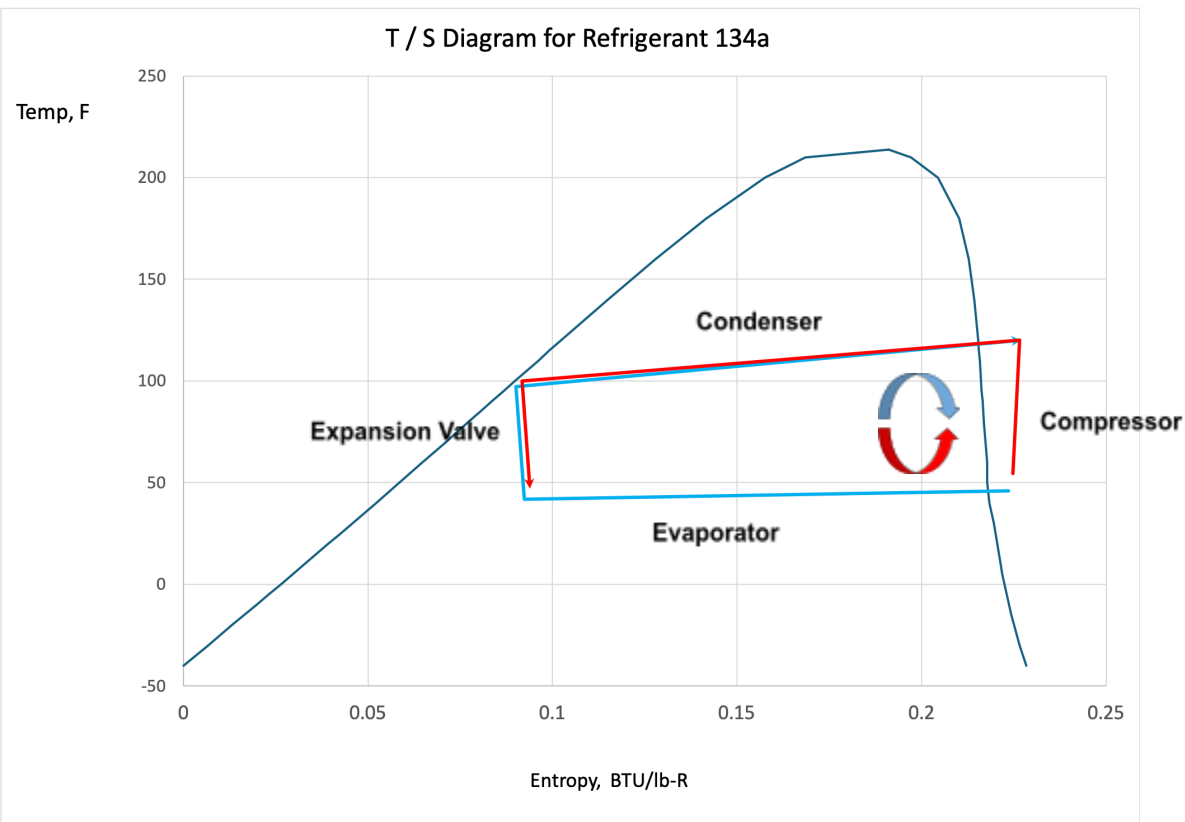
the expansion valve. At its outlet, it was at a temperature of 41.9°F and at the low-side pressure, indicating it was a two-phase mixture, ideal for evaporative heat absorption in our next step.

Table 6 shows the recorded and calculated thermodynamic properties of the refrigerant at each stage of the cycle. Thermodynamic property tables for R134a were used to determine the specific enthalpy and entropy. The expansion valve was assumed to be isenthalpic, allowing for the determination of the temperature at the expansion valve outlet for both cycles. The values were found to be 47.04°F and 41.9°F for the cold and hot cycles respectively.

Airflow measurements confirmed that the evaporator cooled incoming air from 75.3°F to 54.8°F, a 20.5°F drop, indicating substantial heat absorption by the working fluid. On the other hand, the condenser heated air from 72.1°F to 100.8°F, a 28.7°F increase, demonstrating heat rejection by the refrigerant to the indoor air.

Using these temperature changes, airflow, and air properties, we were able to estimate the coefficient of performance (COP). The observed COP was about 3.0, consistent with typical values for heat pumps and confirmation of the advantage vapor-compression heat pumping can offer over electric resistance heating.

Ultimately, the results support theoretical predictions and confirm the energy efficiency and thermodynamic behavior of vapor-compression heat pumps in controlled laboratory settings.



A T-S diagram is shown above, which indicates the states of the refrigerant at each point in the cycle. The heat cycle traverses in a counterclockwise direction, while the refrigeration cycle traverses in a clockwise direction, as shown by the arrow schematic in the middle. The two cycles overlap each other almost perfectly, with errors attributable to real world irreversibilities as well as errors in measurements made.

There are several sources of uncertainty in the heat pump measurement. The first source of potential errors is the calibration. While pressure gauges and thermocouple probes were employed within manufacturers' specifications, slight calibration inaccuracies (particularly when the sensors are bumped or influenced by humidity) can produce systematic departures in measured values. The second source of uncertainty is the transient effect if the system is not in a fully steady state when measurements are made. Though a pause was provided for the readings of non-changing components, slight variations in the compressor speed or air temperatures can produce an underestimation or overestimation of actual operating steady-state conditions.

Another issue is the measurement of airflow with a hand-held anemometer. Minor deviations from alignment between the sensor and the fan inlet air stream or local air turbulence near the inlet might lead to substantial fluctuations. Calculating the volumetric mass flows also assumes a uniform air stream velocity across the cross-section of the fan, which may not always hold well. Such imprecision can propagate through the air-side enthalpy flow calculations and, by extension, into refrigerant mass stream estimations. Pickup or loss of heat outside the main air streams is also possible. For instance, some of the refrigerant lines, the housing of the compressor, or the shells of the condenser/evaporator might conduct heat to or from the laboratory environment. The assumption that "all heat from the condenser is transferred only to the measured airflow" could thus be partly untrue in the event of unmeasured radiation or conduction. Also, table interpolations concerning superheated or saturated R134a data are from idealized properties; if the refrigerant were partly subcooled or a little wet at some points, the direct use of superheated or saturated tables might be incorrect.

Finally, the fan speed adjustment controls the consistency of air flowing through each coil. Changing the fan motor's voltage or mechanical motion can change airflow while in measurement, resulting in temperature or humidity measurement uncertainty. Combining these sources of uncertainty, the calculated Coefficient of Performance (COP) heat transfers likely contain a margin of several percentage points of error. However, the general trends—i.e., the system's capability of delivering substantially more heat than drive work—are likely maintained.

Conclusion

This lab demonstrated the basic principles of a vapor-compression heat pump and its operation with R134a refrigerant. Steady-state low-side pressure of 37 psig and high-side pressure of 129 psig were measured, while the refrigerant temperature at the compressor's discharge was up to around 119 °C. We made heat air-side estimations of heat flows by measuring airflow rates and temperature swings in the evaporator, as in the condenser, and relative humidities. We used the First Law of Thermodynamics in the heat pump's coefficient of performance determination. In line with predictions, a measured COP of around 3.0 demonstrated the system delivered around three units of heat energy per electrical work invested. Though transient response and sensor calibration introduced some uncertainties, the results clearly showed heat pump system efficiency advantages over simple electrical resistive heating. Overall, the exercise reinforced significant thermodynamic principles—especially the nature of phase transformations, change in enthalpy, and work invested—and demonstrated the practicability of the vapor-compression cycle in heating cooler spaces from the heat in warmer spaces.

Appendix

Calculations of enthalpy transfer rates per lb in evaporator and condenser, as well as coefficient of performance for each cycle:

Refer to Table 6 for enthalpy values used in the following calculations

<u>Cold:</u>

$$\frac{dQ_{IN}}{dt} \cdot \frac{1}{\frac{dm}{dt}} = 109.9 - 44.13 = 65.77 \text{ btu/lbm (condenser)}$$

$$\frac{dQ_{OUT}}{dt} \cdot \frac{1}{\frac{dm}{dt}} = 121.1 - 44.13 = 76.97 \text{ btu/lbm (evaporator)}$$

$$\frac{dW_{CYCLE}}{dt} \cdot \frac{1}{\frac{dm}{dt}} = 121.1 - 109.9 = 11.2 \text{ btu/lbm (compressor)}$$

$$COP_R = \frac{\frac{dQ_{IN}}{dt}}{\frac{dW_{CYCLE}}{dt}} = 5.872$$

<u>Hot:</u>

$\frac{dQ_{IN}}{dt} \cdot \frac{1}{\frac{dm}{dt}} =$ 111.4 - 45.12 = 66.28 btu/lbm (evaporator)

$\frac{dQ_{OUT}}{dt} \cdot \frac{1}{\frac{dm}{dt}} =$ 121 - 45.12 = 75.88 btu/lbm (condenser)

$\frac{dW_{CYCLE}}{dt} \cdot \frac{1}{\frac{dm}{dt}} =$ 121 - 111.4 = 9.6 btu/lbm (compressor)

$COP_H = \frac{\frac{dQ_{OUT}}{dt}}{\frac{dW_{CYCLE}}{dt}} =$ 7.904

Calculations of mass flow rates of air and total enthalpy transfer
Refer to tables 3,4,5 for values used in the following calculations

The following specific enthalpies were used for air in the specified state leaving and entering the evaporator/condenser. Values were obtained from thermodynamic tables.

<u>Cold:</u>
Evaporator ~ *Inlet:* **16.01 btu/lbm** *Outlet:* **21.73 btu/lbm** ⇒ Δh = **5.72 btu/lbm**
Condenser ~ *Inlet:* **15.34 btu/lbm** *Outlet:* **10.27 btu/lbm** ⇒ Δh = **-5.07 btu/lbm**

<u>Hot:</u>
Evaporator ~ *Inlet:* **16.15 btu/lbm** *Outlet:* **10.43 btu/lbm** ⇒ Δh = **-5.72 btu/lbm**
Condenser ~ *Inlet:* **15.24 btu/lbm** *Outlet:* **22.78 btu/lbm** ⇒ Δh = **7.54 btu/lbm**

Assuming an air flux area of 55 in^2 and using the outlet airspeeds at the evaporator and condenser, we obtain the following expression for the volume of air per minute:

$$\frac{dV_{air}}{dt}\left(\frac{ft^3}{min}\right) = 55(in^2) \cdot \frac{1}{144}\left(\frac{ft^2}{in^2}\right) \cdot v_{air}\left(\frac{ft}{min}\right)$$

The following volumes of air flow were calculated using the aforementioned expression:

Evaporator out ~ *Cold:* **165.42** $\frac{ft^3}{min}$ *Hot:* **187.95** $\frac{ft^3}{min}$
Condenser out ~ *Cold:* **150.37** $\frac{ft^3}{min}$ *Hot:* **150.37** $\frac{ft^3}{min}$

Taking the density of air to be 0.0765 $\frac{lbm}{ft^3}$, the following mass flow rates of air were found:

Evaporator out ~ *Cold:* **12.65** $\frac{lbm}{min}$ *Hot:* **14.38** $\frac{lbm}{min}$
Condenser out ~ *Cold:* **11.50** $\frac{lbm}{min}$ *Hot:* **11.50** $\frac{lbm}{min}$

Assuming 100% heat transfer between the refrigerant and air, multiplying the aforementioned values by their respective Δh gave the following values for total enthalpy transfer ΔH:

Evaporator ~ *Cold:* **72.36** $\frac{btu}{min}$ *Hot:* **-82.25** $\frac{btu}{min}$
Condenser ~ *Cold:* **-58.3** $\frac{btu}{min}$ *Hot:* **86.71** $\frac{btu}{min}$

Dividing these values by the values for $\frac{dQ_{IN}}{dt} \cdot \frac{1}{\frac{dm}{dt}}$ and $\frac{dQ_{OUT}}{dt} \cdot \frac{1}{\frac{dm}{dt}}$, yielded the following values for the mass flow rates of R134a. The mass flow rate for the cold and hot cycles were taken to be the average of the rates at the evaporator and condenser.

<u>Cold:</u> $\frac{dm}{dt}$= 1.095 lbm/min
<u>Hot:</u> $\frac{dm}{dt}$ = 1.12 lbm/min

Multiplying these values by the intensive heat transfer properties of the cycle yields the following values for the total output for minute of the device in each cycle:

<u>Cold:</u>

$\frac{dQ_{IN}}{dt}$ = 65.77 btu/lbm·1.095 lbm/min = **72.02 btu/min**
$\frac{dQ_{OUT}}{dt}$ = 76.97 btu/lbm ·1.095 lbm/min = **84.28 btu/min**
$\frac{dW_{CYCLE}}{dt}$ = 11.2 btu/lbm ·1.095 lbm/min = **12.26 btu/min**
<u>Hot:</u>

$\frac{dQ_{IN}}{dt}$ = 66.28 btu/lbm ·1.12 lbm/min = **74.23 btu/min**
$\frac{dQ_{OUT}}{dt}$ = 75.88 btu/lbm ·1.12 lbm/min = **84.99 btu/min**
$\frac{dW_{CYCLE}}{dt}$ = 9.6 btu/lbm ·1.12 lbm/min = **10.75 btu/min**